

A Cheatsheet for Lie Algebra

These notes follow Humphreys, *Introduction to Lie Algebras and Representation Theory* [1]. Throughout, F denotes an algebraically closed field.

1 Basic Concepts

1.1 Preliminaries

Definition 1.1. A vector space L over a field F with an operation $L \times L \rightarrow L$ $(x, y) \mapsto [xy]$ called the bracket or commutator of x, y is called a Lie algebra over F if the following axioms are satisfied $\left\{ \begin{array}{l} \text{the bracket operator is bilinear} \\ \forall x \in L, [xx] = 0 \text{ where the last} \\ \forall x, y, z \in L, [x[yz]] + [y[zx]] + [z[xy]] = 0 \end{array} \right.$ condition is called the Jacobi identity.

$$\begin{aligned} [x+y, x+y] &= [x, x] + [x, y] + [y, x] + [y, y] && \text{by the linearity} \\ \rightarrow &= [x, y] + [y, x] && [x, x] = 0, \forall x \in L \Leftrightarrow [x, y] = -[y, x] \\ &= 0 && [x, x] = 0, \forall x \in L \end{aligned}$$

and if $\text{char } F \neq 2$, the converse also holds.

Definition 1.2. Two Lie algebras L, L' over F are isomorphic if there exists a vector space isomorphism $\phi : L \rightarrow L'$ satisfying $\phi([xy]) = [\phi(x)\phi(y)]$, $\forall x, y \in L$.

Definition 1.3. A subalgebra of a Lie algebra L is a subspace K of L s.t. $[xy] \in K$, $\forall x, y \in K$.

$\rightarrow K$ is a Lie algebra with the inherited operations.

e.g. **The General Linear Algebra** $\mathfrak{gl}(V)$

If V is a finite dimensional vector space over F , the set of linear transformation $V \rightarrow V$ is denoted by $\text{End } V$, which is of dimension n^2 where $\dim V = n$ and is a ring relative to the usual product operation. Define a new operation $[x, y] = xy - yx$, called the bracket of x, y . With this operation, $\text{End } V$ becomes a Lie algebra over F .

$\rightarrow \begin{cases} \forall x \in L, [x, x] = xx - xx = 0 \\ \forall x, y, z \in L, [x[yz]] + [y[zx]] + [z[xy]] = x(yz - zy) + y(zx - xz) + z(xy - yx) = 0 \end{cases}$
 This Lie algebra is called the general linear algebra $\mathfrak{gl}(V)$. It is identified with the set of all $n \times n$ matrices over F , denoted $\mathfrak{gl}(n, F)$. Then $[e_{ij}, e_{kl}] = \delta_{jk}e_{il} - \delta_{li}e_{kj}$ where e_{ij} is a standard basis for matrices.

Subalgebras of the General Linear Algebra

- **the Special Linear Algebra** is the set of endomorphisms of V having trace zero and is denoted $\mathfrak{sl}(V) = \mathfrak{sl}(l+1, F)$ where $\dim V = l+1$.

Throughout, $T(\cdot)$ denotes the trace. Since $\begin{cases} T(xy) = T(yx) \\ T(x+y) = T(x) + T(y) \end{cases}$, it is a subalgebra of $\mathfrak{gl}(V)$.

Take all e_{ij} , $i \neq j$ along with $h_i = e_{ii} - e_{i+1, i+1}$. Then this is the basis for $\mathfrak{sl}(V)$ and therefore it has dimension $l + (l+1)^2 - (l+1)$.

- **the symplectic algebra** is the set of endomorphisms x of V satisfying $f(x(v), w) = -f(v, x(w))$, denoted $\mathfrak{sp}(V) = \mathfrak{sp}(2l, F)$, where $\dim V = 2l$ and f has a skew-symmetric form $s = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. Then for $x = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$ where $m, n, p, q \in$

$\mathfrak{gl}(l, F)$, $sx = -x's$ given it being symplectic. In other words, $\begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \begin{bmatrix} m & n \\ p & q \end{bmatrix} =$

$$-\begin{bmatrix} m' & p' \\ n' & q' \end{bmatrix} \Leftrightarrow \begin{cases} n' = n \\ p' = p \\ m' = -q \end{cases}. \text{ Then the basis consists of}$$

$$\begin{cases} m' = -q & \begin{cases} e_{ii} - e_{l+i, l+i}, 1 \leq i \leq l \\ e_{ij} - e_{l+j, l+i} \end{cases} & l + l^2 - l \\ n = n' & \begin{cases} e_{i, l+i} \\ e_{i, l+j} + e_{j, l+i} \end{cases} & l + \frac{1}{2}(l(l-1)) \end{cases}. \text{ Since } n \text{ and } q \text{ share}$$

the basis, the total dimension becomes $2l^2 + l$.

- **the Orthogonal Algebra** is the set of endomorphisms x of V satisfying $f(x(v), w) = -f(v, x(w))$, denoted $\mathfrak{o}(V) = \mathfrak{o}(2l, F)$ where $\dim V = 2l$ and f has a form $s = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$. Then for $x = \begin{bmatrix} m & n \\ p & q \end{bmatrix}$ where $m, n, p, q \in \mathfrak{gl}(l, F)$, $sx = -x's$

given it being orthogonal. In other words, $\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} \begin{bmatrix} m & n \\ p & q \end{bmatrix} = -\begin{bmatrix} m' & p' \\ n' & q' \end{bmatrix} \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}$.

Then the basis consists of

$$\begin{cases} m' = -q & \begin{cases} \text{the diagonal contributes nothing, since } m' = -q \text{ forces } m_{ii} = 0 \\ e_{ij} - e_{l+j, l+i}, i \neq j \end{cases} & l^2 - l \\ n = n' & \begin{cases} e_{i, l+i} \\ e_{i, l+j} + e_{j, l+i} \end{cases} & l + \frac{1}{2}(l(l-1)) \end{cases}$$

Since n and q share the basis, the total dimension becomes $2l^2 - l$.

- $\left\{ \begin{array}{ll} \text{the set of upper triangular matrices } (a_{ij}), a_{ij} = 0, \forall i > j & \mathfrak{t}(n, F) \\ \text{the set of strictly upper triangular matrices } (a_{ij}), a_{ij} = 0, \forall i \geq j & \mathfrak{n}(n, F) \\ \text{the set of all diagonal matrices} & \mathfrak{d}(n, F) \end{array} \right. \rightarrow$
- $\mathfrak{t}(n, F) = \mathfrak{n}(n, F) \oplus \mathfrak{d}(n, F)$
 pf. The basis for $\mathfrak{t}(n, F)$ consists of all e_{ij} s.t. $i \leq j$ which has dimension $\frac{1}{2}n(n+1)$. The basis for $\mathfrak{d}(n, F)$ consists of all e_{ii} which has dimension n , while the basis for $\mathfrak{n}(n, F)$ consists of all e_{ij} s.t. $i < j$ which has dimension $\frac{1}{2}n(n-1)$. Then the sum of bases for $\mathfrak{n}(n, F)$, $\mathfrak{d}(n, F)$ gives the basis for $\mathfrak{t}(n, F)$ and the stmt follows. Moreover the sum is direct: $\mathfrak{n}(n, F)$ is nonabelian while $\mathfrak{d}(n, F)$ is abelian, and their bases are disjoint (e_{ij} with $i < j$ versus e_{ii}), so $\mathfrak{n}(n, F) \cap \mathfrak{d}(n, F) = 0$ and $\mathfrak{t}(n, F) = \mathfrak{n}(n, F) \oplus \mathfrak{d}(n, F)$.

Definition 1.4. For any field F , an F -algebra is a vector space $\mathcal{U}$ over F endowed with a bilinear operation $\mathcal{U} \times \mathcal{U} \rightarrow \mathcal{U}$, usually denoted by juxtaposition. By a derivation of $\mathcal{U}$, we mean a linear map $\delta : \mathcal{U} \rightarrow \mathcal{U}$ satisfying the product rule $\delta(ab) = a\delta(b) + \delta(a)b$.

Since a Lie algebra L is an F -algebra in the above sense, $\text{Der}L$ is defined. If $x \in L$, $y \mapsto [xy]$ is an endomorphism of L , denoted adx . Then $\text{adx} \in \text{Der}L$ by the Jacobi identity. Derivations of this form are called **inner**, all others **outer**.

Definition 1.5. The map $\begin{array}{ccc} L & \rightarrow & \text{Der}L \\ x & \mapsto & \text{adx} \end{array}$ is called the adjoint representation of L .

Definition 1.6. A Lie algebra L over F is called abelian if $\forall x, y \in L$, $[xy] = 0$.

Definition 1.7. A subspace I of a Lie algebra L is called an ideal of L if $x \in L$, $y \in I$ imply $[xy] \in I$.

$\rightarrow [yx] = -[xy] \in I$ by skew-symmetry.

e.g. $\left\{ \begin{array}{l} Z(L) = \{z \in L \mid [xz] = 0, \forall x \in L\}, \text{ the centre of } L \\ [LL] \text{ the derived algebra of } L \end{array} \right.$

Then L is abelian iff $[LL] = 0$. At the other extreme, when $L = \mathfrak{sl}(n, F)$, $[LL] = L$.

If I, J are two ideals of a Lie algebra L , then $\left\{ \begin{array}{l} I + J = \{x + y \mid x \in I, y \in J\} \\ [IJ] = \{\sum [x_i y_i] \mid x_i \in I, y_i \in J\} \end{array} \right.$ are ideals.

Definition 1.8. A Lie algebra L is called simple if it has no ideals except itself and 0, and $[LL] \neq 0$.

e.g. $L = \mathfrak{sl}(2, F)$, $\text{char} F \neq 2$. Then it has a standard basis $x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Then the multiplication table is determined by the equations $\begin{cases} [xy] = h \\ [hx] = 2x \\ [hy] = -2y \end{cases}$. If $I \neq 0$ is an ideal of L , let $ax + by + ch$ be an arbitrary

nonzero element of I . Then applying $\begin{cases} \text{adx twice, we get } -2bx \in I \\ \text{ady twice, we get } -2ay \in I \end{cases}$. If a or b is nonzero, that means either x or y is in I , and therefore $L = I$ by the above multiplication table. If $a = b = 0$, $ch \in I$ and this also implies $L = I$ by the above multiplication table. This shows L is simple.

If L is not simple, it is possible to factor out a nonzero proper ideal I and obtain a **quotient algebra** L/I : as vector space L/I is just the quotient space, while its Lie multiplication is defined by $[x + I, y + I] = [xy] + I$.

Definition 1.9. For any Lie algebras L, L' over F , a linear transformation $\phi : L \rightarrow L'$ is called a *homomorphism* if $\phi([xy]) = [\phi(x)\phi(y)]$, $\forall x, y \in L$. In addition, if $\ker \phi = 0$, ϕ is called a *monomorphism*, or if $\text{Im} \phi = L'$, ϕ is called an *epimorphism*. It is an *isomorphism* if it is both mono- and epi-.

$\ker \phi$ is an ideal of L .

pf. $\forall x \in \ker \phi$, $\forall y \in L$, $\phi([xy]) = [\phi(x)\phi(y)] = 0$ so $[xy] \in \ker \phi$ and therefore the stmt holds.

$\text{Im} \phi$ is a subalgebra of L' .

Proposition 1.1. 1. If $\phi : L \rightarrow L'$ is a homomorphism of Lie algebras, then $L/\ker \phi \cong \text{Im} \phi$. If I is any ideal of L included in $\ker \phi$, there exists a unique

homomorphism $\psi : L/I \rightarrow L'$ making the following diagram commute:

$$\begin{array}{ccc} L & \xrightarrow{\phi} & L' \\ & \searrow \pi & \uparrow \psi \\ & & L/I \end{array}$$

2. If I, J are ideals of L s.t. $I \subset J$, then J/I is an ideal of L/I and $(L/I)/(J/I)$ is naturally isomorphic to L/J .

3. If I, J are ideals of L , there is a natural isomorphism between $(I + J)/J$ and $I/(I \cap J)$.

Definition 1.10. A representation of a Lie algebra L is a homomorphism $\phi : L \rightarrow \mathfrak{gl}(V)$.

For L finite-dimensional, we have an adjoint representation $\text{ad} : L \rightarrow \mathfrak{gl}(L)$. Then its kernel consists of all $x \in L$ for which $\text{adx} = 0$, i.e. $[xy] = 0$. So $\ker \text{ad} = Z(L)$. Then, if L is simple, $Z(L) = 0$ so that $\text{ad}L : L \rightarrow \mathfrak{gl}(L)$ is a monomorphism. This means that any simple Lie algebra is isomorphic to a linear Lie algebra.

Definition 1.11. An automorphism of L is an isomorphism of L onto itself and it is denoted $\text{Aut}L$.

If $g \in \text{GL}(V)$ is any invertible endomorphism of V and if moreover $gLg^{-1} = L$, then $x \mapsto gxg^{-1}$ is an automorphism of L .

Assume $\text{char}F = 0$. Suppose $x \in L$ is an element for which adx is nilpotent, i.e. $(\text{adx})^k = 0$, $\exists k \in \mathbb{Z}_{++}$. Then $\exp(\text{adx}) = 1 + \text{adx} + \frac{(\text{adx})^2}{2!} + \dots + \frac{(\text{adx})^{k-1}}{(k-1)!}$.

CLAIM : $\exp(\text{adx}) \in \text{Aut}L$.

$$\begin{aligned} [\exp \delta(x), \exp \delta(y)] &= \left[\sum_{i=0}^{k-1} \frac{\delta^i x}{i!}, \sum_{j=0}^{k-1} \frac{\delta^j y}{j!} \right] \\ &= \sum_{n=0}^{2k-2} \left(\sum_{i=0}^n \left[\frac{\delta^i x}{i!}, \frac{\delta^{n-i} y}{(n-i)!} \right] \right) \\ \text{pf.} \quad &= \sum_{n=0}^{2k-2} \frac{\delta^n [xy]}{n!} \quad \frac{\delta^n}{n!} [xy] = \sum_{i=0}^n \left[\frac{\delta^i x}{i!}, \frac{\delta^{n-i} y}{(n-i)!} \right] \quad \text{where } \text{adx} \\ &= \sum_{n=0}^{k-1} \frac{\delta^n [xy]}{n!} \quad \delta^k = 0 \\ &= \exp \delta([xy]) \end{aligned}$$

is replaced by an arbitrary nilpotent derivation δ of L . Also given $\exp \delta = 1 + \eta$, its inverse is $1 - \eta + \eta^2 - \dots \pm \eta^{k-1}$ exists and therefore $\exp \delta$ is invertible. This gives the claim.

Definition 1.12. An automorphism of the form $\exp(\text{adx})$, adx nilpotent is called inner and the subgroup of $\text{Aut}L$ generated by these is denoted $\text{Int}L$, the set of inner automorphisms.

$\rightarrow$ it is a normal subgroup ($\phi \in \text{Aut}L$, $x \in L$, $\phi(\text{adx})\phi^{-1} = \text{ad}\phi(x) \rightarrow \phi \exp(\text{adx})\phi^{-1} = \exp(\text{ad}\phi(x))$).

CLAIM : If $L \subset \mathfrak{gl}(V)$ is an arbitrary linear Lie algebra, and $x \in L$ is nilpotent, then $(\exp x)y(\exp x)^{-1} = \exp \text{ad}(x)(y)$, $\forall y \in L$.

pf. $\text{adx} = \lambda_x + \rho_{-x}$ where $\begin{cases} \lambda_x & \text{left multiplication by } x \\ \rho_{-x} & \text{right multiplication by } -x \end{cases} \in \text{End}(\text{End}V)$. Then $\exp \text{adx} = \exp(\lambda_x + \rho_{-x}) = \exp \lambda_x \exp \rho_{-x} = \lambda_{\exp x} \cdot \rho_{\exp -x}$ and the stmt follows.

Definition 1.13. Define a sequence of ideals of L called the derived series by $L^{(0)} = L$, $L^{(1)} = [LL]$, $L^{(2)} = [L^{(1)}L^{(1)}]$, $\dots$, $L^{(i)} = [L^{(i-1)}L^{(i-1)}]$. Then L is said to be solvable if $L^{(n)} = 0$, $\exists n \in \mathbb{Z}_{++}$.

e.g. $\mathfrak{t}(n, F)$ is solvable.

pf. The basis for it consists of the matrix units e_{ij} , $i \leq j$ and the dimension is

$\frac{n(n+1)}{2}$. Since $[e_{ii}e_{il}] = e_{il}$, $\forall i < l$, $\mathfrak{n}(n, F) \subset [LL]$ where it is the subalgebra of upper triangular nilpotent matrices. Since $\mathfrak{t}(n, F) = \mathfrak{d}(n, F) + \mathfrak{n}(n, F)$ and $\mathfrak{d}(n, F)$ is abelian, $\mathfrak{n}(n, F)$ is equal to the derived algebra of L . Define the level of e_{ij} by $j - i$. In the formula of commutators, assume $i < j, k < l$. Wlog, assume $i \neq l$. Then $[e_{ij}e_{kl}] = \begin{cases} e_{il} & j = k \\ 0 & \text{otherwise} \end{cases}$. Each e_{il} is a commutator of two matrices whose levels add up to that of e_{il} . Thus $L^{(2)}$ is spanned by e_{ij} of level ≥ 2 , $L^{(i)}$ by those of level $\geq 2^{i-1}$. Since $L^{(i)} = 0$ whenever $2^{i-1} > n - 1$ and therefore the stmt follows.

Proposition 1.2. *Let L be a Lie algebra.*

1. *If L is solvable, then so are all subalgebras and homomorphic images of L .*
2. *If I is a solvable ideal of L s.t. L/I is solvable, then L itself is solvable.*
3. *If I, J are solvable ideals of L , then so is $I + J$.*

pf.

- If K is a subalgebra of L , $K^{(i)} \subset L^{(i)}$ and therefore the subalgebras are also solvable. If $\phi : L \rightarrow M$ is an epimorphism, an easy induction on i shows $\phi(L^{(i)}) = M^{(i)}$ and therefore its homomorphic images are also solvable.
- Say $(L/I)^{(n)} = 0$. For the canonical homomorphism $\pi : L \rightarrow L/I$, 1 implies $\pi(L^{(n)}) = 0$, i.e. $L^{(n)} \subset I = \ker \pi$. By hypothesis, $\exists m \in \mathbb{Z}_{++}$ s.t. $I^{(m)} = 0 \rightarrow (L^{(n)})^{(m)} = L^{(n+m)} = 0$ so 2 follows.
- There exists an isomorphism between $(I + J)/J$ and $I/(I \cap J)$. As a homomorphic image of I , the RHS is solvable so $(I + J)/J$ is solvable. Then so is $I + J$ by 2.

→let L be an arbitrary Lie algebra and let S be a maximal solvable ideal. If I is any other solvable ideal of L , by the previous proposition 3, $S + I = S$ by maximality, hence $I \subset S$. Thus there exists a unique maximal solvable ideal of L , called the **radical** of L , denoted $\text{Rad}L$. If $\text{Rad}L = 0$, L is called **semisimple**.

1.2 Engel's Theorem

Definition 1.14. *Define a sequence of ideals of L called the descending central series by $L^0 = L, L^1 = [LL], \dots, L^i = [LL^{i-1}]$. L is called **nilpotent** if $\exists n \in \mathbb{Z}_{++}$ s.t. $L^n = 0$.*

→ $L^{(i)} \subset L^i$, $\forall i$ so nilpotent algebras are solvable but not vice versa.

pf. $L = \mathfrak{t}(n, F)$ is solvable. But $L^{(1)} = L^1 = \mathfrak{n}(n, F) = [LL^1] = L^2 = \dots \forall i \geq 1$. Thus it is not nilpotent.

Proposition 1.3. *Let L be a Lie algebra.*

1. *If L is nilpotent, then so are all subalgebras and homomorphic images of L .*
2. *If $L/Z(L)$ is nilpotent, then so is L .*
3. *If L is nilpotent, and nonzero, then $Z(L) \neq 0$.*

pf.

- by the previous proposition, part 1
- $L^n \subset Z(L)$ implies $L^{n+1} = [LL^n] \subset [LZ(L)] = 0$
- for the descending central series, $\exists n \in \mathbb{Z}_{++}$ s.t. $L^{n-1} = [LL^{n-2}] \neq 0, L^n = [LL^{n-1}] = 0$. This means $0 \neq L^{n-1} \subset Z(L)$ and therefore $Z(L) \neq 0$.

$\rightarrow \exists n \in \mathbb{Z}_{++}$ s.t. $\text{adx}_1 \dots \text{adx}_n(y) = 0, \forall x_i, y \in L$ implies L to be nilpotent.

Definition 1.15. *If L is any Lie algebra and $x \in L$, we call x ad-nilpotent if adx is a nilpotent endomorphism.*

Lemma 1.1. *Let $x \in \mathfrak{gl}(V)$ be a nilpotent endomorphism. Then adx is also nilpotent.*

pf. Associate two endomorphisms $\begin{cases} \lambda_x & \text{left translation} \\ \rho_x & \text{right translation} \end{cases}$ to x . Then they are nilpotent since x is. Moreover $\lambda_x \rho_x(y) = xyx = \rho_x \lambda_x(y)$, i.e. they commute. In any ring the sum or difference of two commuting nilpotents is again nilpotent since x, y nilpotent with m, n s.t. $x^m = 0 = y^n, (x \pm y)^{m+n} = 0$. So $\text{adx} = \lambda_x - \rho_x$ is nilpotent.

Theorem 1.1. *Let L be a subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. If L consists of nilpotent endomorphisms and $V \neq 0$, then there exists nonzero $v \in V$ for which $Lv = 0$.*

pf.

- If $\dim L = 0$, the theorem is trivial.
- If $\dim L = 1$, given L is nilpotent $Lv = 0$.
- If $\dim L > 1$, $\exists K \subsetneq L$ proper subalgebra of L . By the previous lemma, K acts via ad as a Lie algebra of nilpotent linear transformation on the vector space L , hence also on the vector space L/K . Because $\dim K < \dim L$, by the inductive hypothesis, $\exists x \in L$ with $x + K \neq K$ such that $x + K$ is killed by the image of K in $\mathfrak{gl}(L/K)$. Then $\forall y \in K, [yx] \in K$ while $x \notin K$, i.e. K is properly

included in $N_L(K)$. Take K to be a maximal proper subalgebra of L . By the previous argument, $N_L(K) = L$, i.e. K is an ideal of L . If $\dim L/K$ were greater than 1, then the inverse image in L of one dimensional subalgebra of L/K would be a proper subalgebra properly containing K , a contradiction. Thus K has codimension 1. Thus $L = K + Fz$, $z \in L \setminus K$. By induction, $W = \{v \in V \mid Kv = 0\}$ is nonzero. Since K is an ideal, W is stable under L , i.e. $\forall x \in L, \forall y \in K, \forall w \in W, yxw = xyw - [xy]w = 0$. Choose $z \in L \setminus K$ as above, so the nilpotent endomorphism z has an eigenvector, i.e. $\exists v \in W$ for which $zv = 0$. Then $Lv = 0$ as desired.

Theorem 1.2 (Engel). *If all elements of L are ad-nilpotent, then L is nilpotent.*

pf. For any Lie algebra L all of whose elements are ad-nilpotent, the algebra $\text{ad}L \subset \mathfrak{gl}(L)$ satisfies the hypothesis of the previous theorem. Thus $\exists 0 \neq x \in L$ s.t. $[Lx] = 0$, i.e. $Z(L) \neq 0$. Now $L/Z(L)$ consists of ad-nilpotent elements and has smaller dimension than L . By inductive hypothesis, $L/Z(L)$ is nilpotent. The previous proposition, part 2, then implies that L itself is nilpotent.

Definition 1.16. *If V is a finite dimensional vector space, a flag in V is a chain of subspaces $0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n = V$, $\dim V_i = i$. If $x \in \text{End}V$ we say x stabilises this flag provided $xV_i \subset V_i$, $\forall i$.*

Corollary 1.1. *Under the hypotheses of the theorem, there exists a flag (V_i) in V stable under L with $xV_i \subset V_{i-1}$, $\forall i$.*

pf. By Theorem 1.0.1, there exists $0 \neq v \in V$ s.t. $Lv = 0$. Set $V_1 = Fv$ and let $W = V/V_1$. Then the induced action of L on W is also by nilpotent endomorphisms. By induction on $\dim V$, W has a flag stabilised by L , whose inverse image in V does the trick.

Lemma 1.2. *Let L be nilpotent, K an ideal of L . Then if $K \neq 0$, $K \cap Z(L) \neq 0$.*

pf. L acts on K via the adjoint representation, so by Theorem 1.0.1, $\exists x \in K$ s.t. $[Lx] = 0$ so $x \in K \cap Z(L)$.

CLAIM : Let L be nilpotent, K a proper subalgebra of L . Then $N_L(K)$ includes K properly.

pf. By Lemma 1.0.1, K acts via ad as a Lie algebra of nilpotent linear transformations on the vector space L , hence also on the vector space L/K . By Theorem 1.0.1, $\exists x' \in L/K$ s.t. $Kx' = 0$. This implies $\forall y \in K, [yx] \in K, x \in L \setminus K$. Thus $x \in N_L(K) \setminus K$ and therefore it is properly contained.

CLAIM : Let L be nilpotent. Then L has an ideal of codimension 1.

pf. Let K be a maximal proper subalgebra of L . By the previous claim, $N_L(K) = L$. Then $[Ly] \subset K$, $\forall y \in K$ and therefore K is an ideal of L . If $\dim L/K$ were greater than 1, the inverse image in L of a one dimensional subalgebra of L/K would be a proper subalgebra properly containing K , a contradiction.

1.3 Theorems of Lie and Cartan

Assume throughout this subsection and the sequel that $\text{char } F = 0$.

Theorem 1.3. *Let L be a solvable subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. If $V \neq 0$, then V contains a common eigenvector for all the endomorphisms in L .*

pf.

- $\dim L = 0$, it holds trivially
- $\dim L = 1$, its proper subalgebra $[LL] = 0$ and L is abelian of dimension 1 and an eigenvector for a basis vector of L gives the theorem.
- $\dim L > 1$, since L is solvable, L properly includes $[LL]$. $L/[LL]$ being abelian, any subspace is automatically an ideal. Take a subspace of codimension 1, then its inverse image K is an ideal of codimension 1 in L . By the inductive hypothesis, $\exists v \in V$, $\exists \lambda : K \rightarrow F$ s.t. $xv = \lambda(x)v$, $\forall x \in K$. Fix this λ and set $W = \{w \in V | xw = \lambda(x)w, \forall x \in K\} \neq 0$.

– **CLAIM : L leaves W invariant.**

pf. $\forall w \in W$, $x \in L$, $\forall y \in K$, $yxw = xyw - [xy]w = \lambda(y)xw - \lambda([xy])w = \lambda(y)(xw)$ iff $\lambda([xy]) = 0$. So we only need to prove $\lambda([xy]) = 0$. Fix $w \in W$, $x \in L$. Let $n \in \mathbb{Z}_{++}$ be the smallest integer for which $w, xw, \dots, x^n w$ are linearly dependent. Let W_i be the subspace of V spanned by $w, xw, \dots, x^{i-1}w$, so $\dim W_n = n$, $W_n = W_{n+i}$, $\forall i \geq 0$ and x maps W_n into W_n . Relative to the basis $w, xw, \dots, x^{n-1}w$ of W_n ,

$$\begin{aligned} yx^i w &= yxx^{i-1}w = xyx^{i-1}w - [xy]x^{i-1}w \\ &= x\lambda(y)x^{i-1}w + xw' - [xy]x^{i-1}w \quad \text{by the inductive hypothesis,} \\ &\equiv \lambda(y)x^i w \pmod{W_i} \quad \text{x maps W_{i-1} into W_i} \end{aligned}$$

where $w' \in W_{i-1}$. This means $y \in K$ is represented by an upper triangular matrix whose diagonal entries are equal to $\lambda(y)$. Then $T|_{W_n}(y) = n\lambda(y)$. In particular, this is true for elements of K of the special form $[xy]$. But x, y both stabilise W_n , so $[xy]$ acts on W_n as the commutator of two endomorphisms of W_n and therefore its trace is 0. So $n\lambda([xy]) = 0$. Given $\text{char } F = 0$, $\lambda([xy]) = 0$.

Then write $L = K + Fz$. Since F is algebraically closed, $\exists v_0 \in W$ an eigenvector of z . Then v_0 is obviously a common eigenvector for L .

Corollary 1.2 (Lie). *Let L be a solvable subalgebra of $\mathfrak{gl}(V)$, $\dim V = n < \infty$. Then L stabilises some flag in V .*

pf. By the previous theorem with induction on $\dim V$.

Corollary 1.3. *Let L be solvable. Then there exists a chain of ideals of L , $0 = L_0 \subsetneq \dots \subsetneq L_n = L$ s.t. $\dim L_i = i$.*

Corollary 1.4. *Let L be solvable. Then $x \in [LL]$ implies $\text{ad}_L x$ is nilpotent. In particular, $[LL]$ is nilpotent.*

pf. Find a flag of ideals in the previous corollary. Relative to a basis $(x_1, \dots, x_n)$ of L for which $(x_1, \dots, x_i)$ spans L_i , the matrices of ad_L lie in $\mathfrak{t}(n, F)$. Thus the matrices of $[\text{ad}_L, \text{ad}_L] = \text{ad}_L[LL] \subset \mathfrak{n}(n, F)$ the derived algebra of $\mathfrak{t}(n, F)$. Thus $\text{ad}_L x$ is nilpotent for $x \in [LL]$. $\text{ad}_{[LL]} x$ is nilpotent, so $[LL]$ is nilpotent by Engel's theorem.

Definition 1.17. $x \in \text{End} V$ is semisimple iff the roots of its minimal polynomial over F are all distinct.

$\Leftrightarrow$ given F being algebraically closed, x is semisimple iff x is diagonalisable.

Proposition 1.4. *Let V be a finite dimensional vector space over F , $x \in \text{End} V$.*

1. *There exists unique $x_s, x_n \in \text{End} V$ satisfying the conditions; $x = x_s + x_n$, $\begin{cases} x_s & \text{semisimple} \\ x_n & \text{nilpotent} \end{cases}$ which commute.*
2. *There exist polynomials $p(T), q(T)$ in one indeterminate, without constant term, s.t. $x_s = p(x)$, $x_n = q(x)$. In particular, x_s, x_n commute with any endomorphism commuting with x .*
3. *If $A \subset B \subset V$ are subspaces, and x maps B into A , then x_s, x_n also map B into A .*

pf. Let $a_1, \dots, a_l$ be the distinct eigenvalues of x , so the characteristic polynomial is $\prod (T - a_i)^{m_i}$ where m_i denote the multiplicity. If $V_i = \ker(x - a_i \cdot 1)^{m_i}$, V is the direct sum of the subspaces $V_1, \dots, V_l$ each stable under x . On V_i , x clearly has characteristic polynomial $(T - a_i)^{m_i}$. By CRT, we get $p(T) \equiv \begin{cases} a_i & \text{mod } (T - a_i)^{m_i} \\ 0 & \text{mod } T \end{cases}$. Set $q(T) = T - p(T)$. Then $p(T), q(T)$ has zero constant term since $p(T) \equiv 0 \pmod T$.

- 2,3

Set $\begin{cases} x_s = p(x) \\ x_n = q(x) \end{cases}$. Since they are polynomials in x , they commute with each other as well as all endomorphisms which commute with x .

They stabilise all subspaces of V stabilised by x , in particular the spaces V_i . The congruence $p(T) \equiv a_i \pmod{(T-a_i)^{m_i}}$ shows the restriction of $x_s - a_i \cdot 1$ to V_i is zero for all i , hence x_s acts diagonally on V_i with single eigenvalue a_i . By definition, $x_n = x - x_s$ which makes it clear that x_n is nilpotent. Because $p(T), q(T)$ have no constant term, 3 follows.

- 1

Existence follows by the previous argument. If there exists $x = s + n$ another decomposition, $x_s - s = n - x_n$. Then by 2, all endomorphisms in the equation commute. Sums of commuting semisimple/nilpotent endomorphisms are semisimple/nilpotent, whereas only 0 can be both semisimple and nilpotent. Thus $x_s = s$, $x_n = n$.

→ the decomposition $x = x_s + x_n$ is called the **Jordan-Chevalley decomposition** of x .

Consider the adjoint representation of the Lie algebra $\mathfrak{gl}(V)$, V finite dimensional. If $x \in \mathfrak{gl}(V)$ is nilpotent, then so is adx by the lemma 1.2.1. Similarly, **if x is semisimple, then so is adx** .

pf. Choose a basis $(v_1, \dots, v_n)$ of V relative to which x has matrix $\text{diag}(a_1, \dots, a_n)$. Let $\{e_{ij}\}$ be the standard basis of $\mathfrak{gl}(V)$ relative to $(v_1, \dots, v_n)$. Then $e_{ij}(v_k) = \delta_{jk}v_i$. Then $\text{adx}(e_{ij}) = (a_i - a_j)e_{ij}$ so adx has diagonal matrix, relative to the chosen basis of $\mathfrak{gl}(V)$.

Lemma 1.3. *Let $x \in \text{End}V$, $x = x_s + x_n$ its Jordan decomposition. Then $\text{adx} = \text{adx}_s + \text{adx}_n$ is the Jordan decomposition of adx .*

pf. Since $\text{adx}_s, \text{adx}_n$ are semisimple and nilpotent respectively, they commute, since $[\text{adx}_s, \text{adx}_n] = \text{ad}[x_s, x_n] = 0$. Thus by the previous proposition 1, the lemma follows.

Lemma 1.4. *Let $\mathcal{U}$ be a finite dimensional F -algebra. Then $\text{Der}\mathcal{U}$ contains the semisimple and nilpotent parts in $\text{End}\mathcal{U}$ of all its elements.*

pf. If $\delta \in \text{Der}\mathcal{U}$, let $\sigma, \nu \in \text{End}\mathcal{U}$ be its semisimple and nilpotent parts respectively. If $a \in F$, set $\mathcal{U}_a = \{x \in \mathcal{U} \mid (\delta - a1)^k x = 0, \exists k\}$. Then $\mathcal{U}$ is the direct sum of those $\mathcal{U}_a$ for which a is an eigenvalue of δ and σ acts on $\mathcal{U}_a$ as scalar multiplication by a . For arbitrary $a, b \in F$, $\mathcal{U}_a \mathcal{U}_b \subset \mathcal{U}_{a+b}$ since $(\delta - (a+b)1)^n(xy) =$

$\sum_{i=0}^n \binom{n}{i} ((\delta - a1)^{n-i}x) \cdot ((\delta - b1)^i y)$, $\forall x, y \in \mathcal{U}$. If $x \in \mathcal{U}_a$, $y \in \mathcal{U}_b$, $\sigma(xy) = (a+b)xy$ because $xy \in \mathcal{U}_{a+b}$. On the other hand, $\sigma xy + x\sigma y = (a+b)xy$. By directness of the sum $\mathcal{U} = \bigoplus \mathcal{U}_a$, $\sigma \in \text{Der } \mathcal{U}$. Then $\nu = \delta - \sigma \in \text{Der } \mathcal{U}$.

Lemma 1.5. *Let $A \subset B$ be two subspaces of $\mathfrak{gl}(V)$, $\dim V < \infty$. Set $M = \{x \in \mathfrak{gl}(V) \mid [xB] \subset A\}$. Suppose $x \in M$ satisfies $T(xy) = 0$, $\forall y \in M$. Then x is nilpotent.*

pf. Let $x = s + n$ be the Jordan decomposition of x . Fix a basis $v_1, \dots, v_m$ of V relative to which s has matrix $\text{diag}(a_1, \dots, a_m)$. Let E be the vector subspace of F over the prime field $\mathbb{Q}$ spanned by the eigenvalues $a_1, \dots, a_m$. Consider the dual space E^* , whose elements are the linear functions $f : E \rightarrow \mathbb{Q}$. Given such $f \in E^*$, let y be any element of $\mathfrak{gl}(V)$ whose matrix relative to the given basis is $\text{diag}(f(a_1), \dots, f(a_m))$. If $\{e_{ij}\}$ is the corresponding basis of $\mathfrak{gl}(V)$, $\begin{cases} \text{ads}(e_{ij}) = (a_i - a_j)e_{ij} \\ \text{ady}(e_{ij}) = (f(a_i) - f(a_j))e_{ij} \end{cases}$. Let $r(T) \in F[T]$ be a polynomial without constant term satisfying $r(a_i - a_j) = f(a_i) - f(a_j)$ for all pairs i, j . The existence of such $r(T)$ follows from Lagrange interpolation. Then $\text{ady} = r(\text{ads})$. Since ads is the semisimple part of adx by the lemma 1.3.1, it can be written as a polynomial in adx without constant term by the proposition 1.3.1. Therefore $\text{ady} = r(\text{ads})$ is also a polynomial in adx without constant term. By the hypothesis of the lemma, adx maps B into A , so $\text{ady}(B) \subset A$ i.e. $y \in M$. By the hypothesis of the lemma, $T(xy) = 0 \rightarrow \sum a_i f(a_i) = 0$. The LHS is a $\mathbb{Q}$ -linear combination of elements of E . Applying f to the above equation gives $\sum (f(a_i))^2 = 0$. But the numbers $f(a_i)$ are rational, so this forces all of them to be 0, and therefore f must be 0 because the a_i span E . Then E^* is zero and its dual E is also 0 and the lemma follows.

Lemma 1.6. *$T([xy]z) = T(x[yz])$ for all endomorphisms x, y, z of a finite-dimensional vector space.*

pf. $\begin{cases} [xy]z = xyz - yxz \rightarrow T([xy]z) = T(xyz) - T(yxz) \\ x[yz] = xyz - xzy \rightarrow T(x[yz]) = T(xyz) - T(xzy) \end{cases}$ are equivalent given $T(ab) = T(ba)$.

Theorem 1.4 (Cartan's Criterion). *Let L be a subalgebra of $\mathfrak{gl}(V)$, V finite dimensional. Suppose that $T(xy) = 0$, $\forall x \in [LL]$, $y \in L$. Then L is solvable.*

pf. Let $\begin{cases} A = [LL] \\ B = L \end{cases}$, so $M = \{x \in \mathfrak{gl}(V) \mid [xL] \subset [LL]\}$. Then $L \subset M$ trivially. By the hypothesis $T(xy) = 0$, $\forall x \in [LL]$, $\forall y \in L$. If $[xy]$ is a typical generator of $[LL]$, and if $z \in M$, the equation above shows $T([xy]z) = T(x[yz]) = T([yz]x)$. By the definition of M , $[yz] \in [LL]$ so $T(xy) = 0$, $\forall x \in [LL]$, $\forall y \in M$. Then by the previous lemma, every $x \in [LL]$ is nilpotent. Then by Engel's theorem, $[LL]$ is nilpotent and therefore L is solvable.

2 Semisimple Algebras

2.1 Killing Form

Definition 2.1. Let L be any Lie algebra. If $x, y \in L$, define the killing form $\kappa(x, y) = T(\text{adxady})$.

→ it is symmetric bilinear and associative in the sense that $\kappa([xy], z) = \kappa(x, [yz])$.

Lemma 2.1. Let I be an ideal of L . If κ is the killing form of L and κ_I be the killing form of I , then $\kappa_I = \kappa|_{I \times I}$.

pf. If W is a subspace of a finite dimensional vector space V , and ϕ an endomorphism of V mapping V into W , then $T\phi = T\phi|_W$, since we can extend a basis of W to a basis of V and calculate ϕ . Then if $x, y \in I$, adxady is an endomorphism of L , mapping L into I , so its trace $\kappa(x, y)$ coincides with the trace $\kappa_I(x, y)$ of $\text{adxady}|_I = \text{ad}_I x \text{ad}_I y$.

Definition 2.2. A symmetric bilinear form $\beta(x, y)$ is called nondegenerate if its radical S is 0, where $S = \{x \in L | \beta(x, y) = 0, \forall y \in L\}$.

→ S is an ideal of L , given the associativity of the killing form.

Theorem 2.1. Let L be a Lie algebra. Then L is semisimple iff its killing form is nondegenerate.

pf.

- Assume L is semisimple. Then $\text{Rad}L = 0$. Let S be the radical of κ . By definition, $T(\text{adxady}) = 0, \forall x \in S, \forall y \in L$. By the Cartan criterion, $\text{ad}_L S$ is solvable, hence S is solvable. Since S is an ideal of L , $S \subset \text{Rad}L = 0$ and κ is nondegenerate.
- Let $S = 0$ and I be an abelian ideal of L . Suppose $x \in I, y \in L$. Then adxady maps $L \rightarrow L \rightarrow I$ and $(\text{adxady})^2$ maps L into $[II] = 0$. This means adxady is nilpotent, hence $0 = T(\text{adxady}) = \kappa(x, y)$ so $I \subset S = 0$.

→ $S \subset \text{Rad}L$ but not vice versa.

pf. Let L be a Lie algebra with basis x, y, z s.t.
$$\begin{cases} [xy] = z \\ [xz] = -y \\ [yz] = 0 \end{cases} \text{ . Then } \text{adx} =$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \text{ady} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \text{ad}(z) = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and therefore its } \kappa = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

So $\text{Rad}L = L \not\subset \text{span}(y, z) = S \equiv \text{radical of } \kappa$.

Definition 2.3. A Lie algebra L is said to be the direct sum of ideals $I_1, \dots, I_t$ provided $L = I_1 + \dots + I_t$, denoted $L = I_1 \oplus \dots \oplus I_t$.

$$\rightarrow [I_i I_j] \subset I_i \cap I_j = 0, \forall i \neq j.$$

Theorem 2.2. Let L be semisimple. Then there exist ideals $L_1, \dots, L_t$ of L which are simple s.t. $L = L_1 \oplus \dots \oplus L_t$. Every simple ideal of L coincides with one of the L_i . Moreover, the killing form of L_i is the restriction of κ to $L_i \times L_i$.

pf.

- **CLAIM : Let I be an arbitrary ideal of L . Then $L = I \oplus I^\perp$.**

pf. $I^\perp = \{x \in L : \kappa(x, y) = 0, \forall y \in I\}$ is also an ideal by the associativity of κ . By Cartan's criterion applied to $I \cap I^\perp$, it is solvable, hence 0 as an ideal of the semisimple L . Thus since $\dim I + \dim I^\perp = \dim L$, $L = I \oplus I^\perp$.

- Existence

If L has no proper nonzero ideal, then L itself is simple and there is nothing to prove. Otherwise, let L_1 be a minimal nonzero ideal. Then by the first claim, $L = L_1 \oplus L_1^\perp$. In particular, any ideal of L_1 is also an ideal of L and therefore L_1 is semisimple, and by minimality it is simple. For the same reason $L_1^\perp$ is semisimple. By inductive hypothesis, it splits into a direct sum of simple ideals, which are also ideals of L .

- Uniqueness

If I is any simple ideal of L , then $[IL]$ is also an ideal of L , nonzero because $Z(L) = 0$. Then $[IL] = I$ by simplicity. On the other hand, $[IL] = [IL_1] + \dots + [IL_t]$ so all but one summand must be 0. Say $[IL_i] = I$. Then $I \subset L_i$, which forces $I = L_i$ by the simplicity of L_i .

Corollary 2.1. If L is semisimple, $L = [LL]$ and all ideals and homomorphic images of L are semisimple. Moreover, each ideal of L is a sum of certain simple ideals of L .

For any Lie algebra L , $\text{ad}L$ is an ideal in $\text{Der}L$.

pf. $\forall y, [\delta, \text{adx}](y) = \delta \text{adx}(y) - \text{adx} \delta(y) = \delta([xy]) - [x, \delta y] = [\delta x, y] + [x, \delta y] - [x, \delta y] = [\delta x, y] = \text{ad}(\delta x)(y) \Leftrightarrow [\delta, \text{adx}] = \text{ad}(\delta x), \forall x \in L, \forall \delta \in \text{Der}L$. So the stmt follows.

Theorem 2.3. If L is semisimple, then $\text{ad}L = \text{Der}L$.

pf. Since L is semisimple, $Z(L) = 0$. Thus $L \rightarrow \text{ad}L$ is an isomorphism of Lie algebras. In particular, $M = \text{ad}L$ has nondegenerate killing form by the theorem 1.4.1. If $D = \text{Der}L$, $[DM] \subset M$. Then by the lemma 1.4.1, κ_M is the restriction to $M \times M$ of the killing form κ_D of D . In particular, if $I = M^\perp$ is the subspace of D orthogonal to M under κ_D , then the nondegeneracy of κ_M forces $I \cap M = 0$. Both I, M are ideals of D so $[IM] = 0$. If $\delta \in I$, $\text{ad}\delta x = 0$, $\forall x \in L$ by the previous stmt, so $\delta x = 0$, $\forall x \in L$ because ad is one to one and $\delta = 0$. So $I = 0$, $\text{Der}L = M = \text{ad}L$.

→ For any semisimple Lie algebra L , if $\mathcal{U}$ is any F -algebra of finite dimension, $\text{Der}\mathcal{U}$ contains the semisimple and nilpotent parts in $\text{End}\mathcal{U}$ of all its elements. In particular, since $\text{Der}L$ coincides with $\text{ad}L$ while $L \rightarrow \text{ad}L$ is one to one, each $x \in L$ determines unique elements $s, n \in L$ s.t. $\text{ad}x = \text{ad}s + \text{ad}n$ is the Jordan decomposition of $\text{ad}x$. This means $x = s + n$, $[sn] = 0$, $\begin{cases} s & \text{ad-semisimple} \\ n & \text{ad-nilpotent} \end{cases}$ and $\begin{cases} s = x_s & \text{the semisimple part} \\ n = x_n & \text{the nilpotent part} \end{cases}$ of x .

2.2 Complete Reducibility of Representations

Definition 2.4. Let L be a Lie algebra. A vector space V , endowed with an operation $\begin{matrix} L \times V & \rightarrow & V \\ (x, v) & \mapsto & xv \end{matrix}$ is called an L -module if the following conditions are satisfied: $\begin{cases} (ax + by)v = a(xv) + b(yv), x(av + bw) = a(xv) + b(xw) \\ [xy]v = xyv - yxv \end{cases}, \forall x, y \in L. \forall v, w \in V, a, b \in F.$

e.g. If $\phi : L \rightarrow \text{gl}(V)$ is a representation of L , then V may be viewed as an L -module via the action $xv = \phi(x)(v)$. Conversely, given an L -module V , this equation defines a representation $\phi : L \rightarrow \text{gl}(V)$.

Definition 2.5. A homomorphism of L -modules is a linear map $\phi : V \rightarrow W$ s.t. $\phi(xv) = x\phi(v)$. When ϕ is an isomorphism of vector spaces, we call it an isomorphism of L -modules.

Definition 2.6. An L -module V is called irreducible if it has precisely two L -submodules- itself and 0 .

Definition 2.7. V is called completely reducible if V is a direct sum of irreducible L -submodules or equivalently if each L -submodule W of V has a complement W' .

→ if it is a direct sum of irreducible L -submodules, its any submodule W is also a direct sum of irreducible L -submodules. Then the summand that does not belong to its submodule also makes a L submodule $W^\perp$ s.t. $W \cap W^\perp = 0$, $W + W^\perp = L$

so $L = W \oplus W^\perp$. Conversely, if any submodule of L has its complement, pick a minimal submodule of L . If there is no proper submodule, L is irreducible and there is nothing to prove. Otherwise, pick a minimal proper submodule L_1 of L . Then by minimality, it is irreducible. Then it has complement $L_1^\perp$ with $\dim L_1^\perp < \dim L$. Then by the inductive hypothesis it is a direct sum of irreducible L -submodules and therefore $L = L_1 \oplus L_1^\perp$ is.

Given a representation $\phi : L \rightarrow \mathfrak{gl}(V)$, the associative algebra generated by $\phi(L)$ in $\text{End}V$ leaves invariant precisely the same subspaces as L . Therefore, all the usual results for modules over associative rings hold for L as well.

SCHUR'S LEMMA : Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be irreducible. Then the only endomorphisms of V commuting with all $\phi(x)$ are the scalars.

L itself is an L -module, and then an L -submodule is just an ideal, so a simple algebra L is irreducible as L -module, while a semisimple algebra is completely reducible.

MAKING NEW MODULES out of OLD ONES

- dual module

Let V be a L -module. Then the dual vector space V^* becomes an L -module with $(xf)(v) = -f(xv)$, $\forall f \in V^*$. Then $([xy]f)(v) = -f([xy]v) = -f(xyv - yxv) = -f(xyv) + f(yxv) = (xf)(yv) - (yf)(xv) = -(yx f)(v) + (xy f)(v) = ((xy - yx)f)(v)$.

- tensor-product

If V, W are L -modules, let $V \otimes W$ be the tensor product over F of the underlying vector spaces. For Lie algebras, define $x(v \otimes w) = xv \otimes w + v \otimes xw$. Then $[xy](v \otimes w) = [xy]v \otimes w + v \otimes [xy]w = (xyv - yxv) \otimes w + v \otimes (xyw - yxw) = (xyv \otimes w + v \otimes xyw) - (yxv \otimes w + v \otimes yxw) = (xy - yx)(v \otimes w)$.

→ given a vector space V over F , $\exists$ an isomorphism
$$\begin{array}{ccc} V^* \otimes V & \rightarrow & \text{End}V \\ f \otimes v & \mapsto & (w \mapsto f(w)v) \end{array}$$
 enables us to consider $\text{End}V$ as an L -module via the isomorphism. The action of L on $\text{End}V$ is $(x \cdot \psi)(v) = x\psi(v) - \psi(xv)$, $\forall \psi \in \text{End}V$, $\forall v \in V$.

Let L be semisimple and $\phi : L \rightarrow \mathfrak{gl}(V)$ be a faithful representation of L . Define a symmetric bilinear form $\beta(x, y) = T(\phi(x)\phi(y))$ on L . The form β is associative, so its radical S is an ideal of L . Moreover $\phi(S) \cong S$ is solvable, so $S = 0$ and β is non-degenerate. Let L be semisimple, β any nondegenerate symmetric associative bilinear form on L . If $(x_1, \dots, x_n)$ is a basis of L , there is a uniquely determined dual basis $(y_1, \dots, y_n)$ relative to β satisfying $\beta(x_i, y_j) = \delta_{ij}$. If $x \in L$,
$$\begin{cases} [xx_i] = \sum_j a_{ij} x_j \\ [xy_i] = \sum_j b_{ij} y_j \end{cases}.$$

$$\begin{aligned}
a_{ik} &= \sum_j a_{ij} \beta(x_j, y_k) = \beta([xx_i], y_k) && \text{by the definition of } \beta \\
\text{Then} \quad &= \beta(-[x_i x], y_k) && \text{by the skew-symmetry of bracket} \\
&= \beta(x_i, -[xy_k]) && \text{by the associativity of } \beta \\
&= -\sum_k b_{kj} \beta(x_i, y_j) && = -b_{ki}
\end{aligned}$$

$L \rightarrow \mathfrak{gl}(V)$ is any representation of L , write $c_\phi(\beta) = \sum_i \phi(x_i) \phi(y_i) \in \text{End} V$ where

$$\begin{aligned}
[\phi(x) c_\phi(\beta)] &= \sum_i [\phi(x) \phi(x_i)] \phi(y_i) + \sum_i \phi(x_i) [\phi(x) \phi(y_i)] \\
x_i, y_i \text{ run over dual bases relative to } \beta. \text{ Then} &= \sum_{ij} a_{ij} \phi(x_j) \phi(y_i) + \sum_{ij} b_{ij} \phi(x_i) \phi(y_j) \\
&= 0
\end{aligned}$$

Thus $c_\phi(\beta)$ is an endomorphism of V commuting with $\phi(L)$.

$\rightarrow c_\phi(\beta)$ is called the Casimir element of ϕ and its trace is $\sum_i T(\phi(x_i) \phi(y_i)) = \sum_i \beta(x_i, y_i) = \dim L$. If ϕ is also irreducible, by Schur's lemma, it should be scalar.

e.g. $L = \mathfrak{sl}(2, F)$, $V = F^2$, ϕ is the identity map $L \rightarrow \mathfrak{gl}(V)$. Let (x, h, y) be the standard basis of L . Then its dual basis relative to the trace form is $(y, h/2, x)$ so

$$c_\phi = xy + \frac{h^2}{2} + yx = \begin{bmatrix} 3/2 & 0 \\ 0 & 3/2 \end{bmatrix}.$$

When ϕ is not faithful, we can factor out $\ker \phi$ and get L' the sum of the remaining simple ideals after factoring out $\ker \phi$. Then $\phi|_{L'}$ is faithful and we can get the Casimir element for L' . Then $\phi(L) = \phi(L')$ commute with the Casimir element.

Lemma 2.2. *Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be a representation of a semisimple Lie algebra L . Then $\phi(L) \subset \mathfrak{sl}(V)$. In particular, L acts trivially on any one dimensional L -module.*

pf. $L = [LL]$ and $\mathfrak{sl}(V)$ is the derived algebra of $\mathfrak{gl}(V)$, the lemma follows.

Theorem 2.4 (Weyl). *Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be a finite dimensional representation of a semisimple Lie algebra. Then ϕ is completely reducible.*

pf.

- V has an L -submodule W of codimension 1

Since by the previous lemma, L acts trivially on V/W , wlog we may denote this module F and get a s.e.s. $0 \rightarrow W \rightarrow V \rightarrow F \rightarrow 0$.

– **CLAIM : wlog we can assume W is an irreducible L -module.**

Let W' be a proper nonzero submodule of W . Then we get another s.e.s. $0 \rightarrow W/W' \rightarrow V/W' \rightarrow F \rightarrow 0$. By the inductive hypothesis, it splits, so there is a one dimensional L -submodule of V/W' complementary to W/W' . Let it be $\tilde{W}/W'$. Then there exists another s.e.s $0 \rightarrow$

$W' \rightarrow \tilde{W} \rightarrow F \rightarrow 0$ with $\dim W' < \dim W$, so by the inductive hypothesis, there exists a submodule X complementary to W' , i.e. $\tilde{W} = W' \oplus X$. But $V/W' = W/W' \oplus \tilde{W}/W'$ so $V = W \oplus X$, given $W \cap X = 0$ and dimension matching.

Let $c = c_\phi$ be the Casimir element of ϕ . Since c commutes with $\phi(L)$, c is actually an L -module endomorphism of V . In particular, $c(W) \subset W$ and $\ker c$ is an L -submodule of V . Because L acts trivially on V/W , c must do likewise as a linear combination of products of elements $\phi(x)$. So c has trace 0 on V/W . On the other hand, c acts as a scalar on the irreducible L -submodule W by the Schur's lemma. This cannot be 0, because that would force $T|_V(c) = 0$, a contradiction. Thus $\ker c$ is a one-dimensional L -submodule of V which intersects W trivially, which is the desired complement of W .

- General case

Let W be a nonzero submodule of V . Then there exists a s.e.s. $0 \rightarrow W \rightarrow V \rightarrow V/W \rightarrow 0$. Let $\text{hom}(V, W)$ be the space of linear maps $V \rightarrow W$, viewed as L -module. Let $\mathcal{V}$ be the subspace of $\text{hom}(V, W)$ consisting of those maps whose restriction to W is a scalar multiplication. Then $f|_W = a1_W, \forall x \in L, \forall w \in W, (xf)(w) = xf(w) - f(xw) = a(xw) - a(xw) = 0 \rightarrow (xf)|_W = 0$ so $\mathcal{V}$ is an L -submodule. Let $\mathcal{W}$ be the subspace of $\mathcal{V}$ consisting of those f whose restriction to W is zero. Then $\mathcal{W}$ is also an L -submodule and L maps $\mathcal{V}$ to $\mathcal{W}$. Moreover $\mathcal{V}/\mathcal{W}$ has dimension one because $f \in \mathcal{V}$ is determined by the scalar $f|_W \bmod \mathcal{W}$. This gives the first situation. Then $\mathcal{V}$ has a one-dimensional submodule complementary to $\mathcal{W}$. Let $f : V \rightarrow W$ span it, so after multiplying by a nonzero scalar, we may assume $f|_W = 1_W$. Since f maps V into W and acts as 1_W on W , we can say $V = W \oplus \ker f$.

Theorem 2.5. *Let $L \subset \mathfrak{gl}(V)$ be a semisimple linear Lie algebra with V finite dimensional. Then L contains the semisimple and nilpotent parts in $\mathfrak{gl}(V)$ of all its elements. In particular, the abstract and usual Jordan decompositions in L coincide.*

pf. Let $x \in L$ arbitrary, with Jordan decomposition $x = x_s + x_n$ in $\mathfrak{gl}(V)$. Since $\text{adx}(L) \subset L$, $\begin{cases} \text{adx}_s(L) \subset L \\ \text{adx}_n(L) \subset L \end{cases}$ where $\text{ad} = \text{ad}_{\mathfrak{gl}(V)}$, i.e. $x_s, x_n \in N_{\mathfrak{gl}(V)}(L) = N$, which is a Lie subalgebra of $\mathfrak{gl}(V)$ including L as an ideal. If W is any L -submodule of V , define $L_W = \{y \in \mathfrak{gl}(V) | y(W) \subset W, T(y|_W) = 0\}$. For example, $L_V = \mathfrak{sl}(V)$. Since $L = [LL]$, L lies in all such L_W . Set $L' =$ intersection of N with all spaces L_W . Then L' is a subalgebra of N including L as an ideal. Then if $x \in L$, x_s, x_n also lie in L_W and therefore in L' .

- **CLAIM :** $L = L'$.

pf. L' being finite dimensional L -module, by Weyl's theorem $L' = L + M$ for some L -submodule M , where the sum is direct. But $[LL'] \subset L$ given $L' \subset L$ so the action of L on M is trivial. Let W be any irreducible L -submodule of V . If $y \in M$, $[Ly] = 0$ so by Schur's lemma, y acts on W as a scalar. On the other hand, $T(y|_W) = 0$ because $y \in L_W$. Therefore y acts on W as zero. V can be written as a direct sum of irreducible L -submodules by Weyl's theorem so $y = 0$. Thus $M = 0$, $L = L'$.

Corollary 2.2. *Let L be a semisimple Lie algebra, $\phi : L \rightarrow \mathfrak{gl}(V)$ a finite dimensional representation of L . If $x = s + n$ is the abstract Jordan decomposition of $x \in L$, $\phi(x) = \phi(s) + \phi(n)$ is the usual Jordan decomposition of $\phi(x)$.*

pf. The algebra $\phi(L)$ is spanned by the eigenvectors of $\text{ad}_{\phi(L)}\phi(s)$ since L has this property relative to ads , and therefore $\text{ad}_{\phi(L)}\phi(s)$ is semisimple. Similarly, $\text{ad}_{\phi(L)}\phi(n)$ is nilpotent and it commutes with $\text{ad}_{\phi(L)}\phi(s)$. Thus $\phi(x) = \phi(s) + \phi(n)$ is the abstract Jordan decomposition of $\phi(x)$ in the semisimple Lie algebra $\phi(L)$. Then by the previous theorem, the corollary follows.

Definition 2.8. *A Lie algebra L for which $\text{Rad}L = Z(L)$ is called reductive.*

e.g. $\left\{ \begin{array}{l} L \text{ abelian} \\ L = \mathfrak{gl}(n, F) \end{array} \right\}$, then it is reductive.

- **If L is reductive, L is completely reducible $\text{ad}L$ -module.**

pf. By the hypothesis $L/Z(L)$ is semisimple. Since $\text{ad}L$ acts trivially on $Z(L)$, the adjoint representation of L on L factors through $L/Z(L)$. Thus L is an $L/Z(L)$ -module. $\phi : L/Z(L) \rightarrow \mathfrak{gl}(V)$ is a representation of a semisimple Lie algebra. Then ϕ is completely reducible by the Weyl's theorem. Thus L is completely reducible.

- **If L is a completely reducible $\text{ad}L$ -module, then L is reductive.**

pf. There is an adjoint representation $\phi : L \rightarrow \mathfrak{gl}(V)$, let $R = \text{Rad}L$ be its radical. Then since it is an ideal of L , it is an $\text{ad}L$ -submodule of L . Then, $L = R \oplus S$ by L being completely reducible. Because $\text{ad}L$ -submodules of the adjoint module are ideals, S is also an ideal. Thus $\left\{ \begin{array}{l} [RS] \subset R \\ [RS] \subseteq S \end{array} \right.$. Thus $[RS] \subseteq R \cap S = 0 \rightarrow [RS] = 0$. Since R is solvable, $[RR] \subsetneq R$ unless $R = 0$ and $[RR]$ is also an $\text{ad}L$ -submodule. By complete reducibility, choose an $\text{ad}L$ -submodule $T \subseteq R$ s.t. $R = [RR] \oplus T$. Given T an ideal, $\left\{ \begin{array}{l} [RT] \subseteq T \\ [RT] \subseteq [RR] \end{array} \right. \rightarrow$

$[RT] = 0$, $R = [RR] \oplus T$. Thus $[RR] = [R[RR]]$. If $[RR] \neq 0$, $R^{(1)} = R^{(2)} = \dots$, a contradiction. Thus $[RR] = 0$ and $R \subseteq Z(L)$. Since $Z(L) \subseteq R$ the stmt follows.

- **If L is reductive, then all finite dimensional representations of L in which $Z(L)$ is represented by semisimple endomorphisms are completely reducible.**

pf. By the first stmt, it is a completely reducible $\text{ad}L$ -module and $L = Z(L) \oplus [LL]$ where $[LL]$ is semisimple. Let $\phi : L \rightarrow \mathfrak{gl}(V)$ be finite dimensional with every element of $Z(L)$ represented semisimply. By Weyl's theorem, it is completely reducible as $[LL]$ -module. $[Z(L), L] = 0$ implies $[\phi(z)\phi(x)] = 0$, $\forall z \in Z(L)$, $\forall x \in [LL]$. Since $\phi(z)$ are commuting semisimple endomorphisms over an algebraically closed field they are simultaneously diagonalisable. Thus $V = \bigoplus_{\lambda} V_{\lambda}$ where $\forall V_{\lambda}$ is a common eigenspace for $Z(L)$. Since $\phi(z)$ commute with $\phi([LL])$, $xV_{\lambda} \subseteq V_{\lambda}$, $\forall x \in [LL]$. So V_{λ} is an $[LL]$ -module and by Weyl's theorem $V_{\lambda} = \bigoplus_j V_{\lambda,j}$ where $V_{\lambda,j}$ are irreducible $[LL]$ -module. But $Z(L)$ acts on all $V_{\lambda,j}$ by the scalar $\lambda(z)$, $V_{\lambda,j}$ is actually an L -submodule. If it is a proper L -submodule, then it is an $[LL]$ -submodule, contradicting irreducibility. Hence $V = \bigoplus_{\lambda,j} V_{\lambda,j}$ is a decomposition into irreducible L -modules.

2.3 Representations of $\mathfrak{sl}(2, F)$

Let L denote $\mathfrak{sl}(2, F)$ whose standard basis consists of $x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $h = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Then $[hx] = 2x$, $[hy] = -2y$, $[xy] = h$.

Let V be an arbitrary L -module. Since h is semisimple, h acts diagonally on V by the previous corollary. Then V is decomposed as direct sum of eigenspaces $V_{\lambda} = \{v \in V | hv = \lambda v\}$, $\lambda \in F$. Whenever $V_{\lambda} \neq 0$, λ is called a weight of h in V and we call V_{λ} a weight space.

Lemma 2.3. If $v \in V_{\lambda}$, $\begin{cases} xv \in V_{\lambda+2} \\ yv \in V_{\lambda-2} \end{cases}$.

pf. $\begin{cases} h(xv) = [hx]v + xhv = 2xv + \lambda xv = (\lambda + 2)xv \\ h(yv) = [hy]v + yhv = -2yv + \lambda yv = (\lambda - 2)yv \end{cases}$
 $\rightarrow x, y$ are represented by nilpotent endomorphisms of V .

Since $\dim V < \infty$ and $V = \bigoplus_{\lambda \in F} V_{\lambda}$ is direct, there must exist $V_{\lambda} \neq 0$ s.t. $V_{\lambda+2} = 0$. For such λ , any nonzero vector in V_{λ} will be called a maximal vector of weight

λ .

Assume V is an irreducible L -module. Choose a maximal vector, say $v_0 \in V$ and set $v_{-1} = 0$, $v_i = \frac{1}{i!}y^i v_0$, $\forall i \geq 0$.

Lemma 2.4. $\begin{cases} (i) & hv_i = (\lambda - 2i)v_i \\ (ii) & yv_i = (i+1)v_{i+1} \\ (iii) & xv_i = (\lambda - i + 1)v_{i-1} \end{cases}$

pf.

- for $i = 0$, $hv_0 = \lambda v_0$
- $i > 0$, $hv_i = \frac{1}{i}hyv_{i-1} = \frac{1}{i}([hy]v_{i-1} + yhv_{i-1}) = \frac{1}{i}(-2yv_{i-1} + (\lambda - 2(i-1))yv_{i-1}) = \frac{1}{i}((\lambda - 2i)yv_{i-1}) = (\lambda - 2i)v_i$
- $(i+1)v_{i+1} = (i+1)\frac{1}{(i+1)!}y^{i+1}v_0 = yv_i$
- $i = 0 \rightarrow xv_0 = (\lambda + 1)v_{-1} = (\lambda + 1)0 = 0$
 $ixv_i = xyv_{i-1} = [xy]v_{i-1} + yxv_{i-1}$ definition of bracket
 $= hv_{i-1} + yxv_{i-1}$
 $i > 0 \rightarrow = (\lambda - 2(i-1))v_{i-1} + (\lambda - i + 2)yv_{i-2}$ i and inductive hypothesis
 $= (\lambda - 2i + 2)v_{i-1} + (i-1)(\lambda - i + 2)v_{i-1}$ by (ii)
 $= i(\lambda - i + 1)v_{i-1}$

$\rightarrow$ (i) implies the nonzero v_i are all linearly independent. But $\dim V < \infty$. Let m be the smallest integer for which $\begin{cases} v_m \neq 0 \\ v_{m+1} = 0 \end{cases}$. Then the subspace of V with basis $(v_0, \dots, v_m)$ is an L -submodule different from 0. Because V is irreducible, this subspace must be all of V and moreover relative to the ordered basis $(v_0, \dots, v_m)$, the matrices of the endomorphisms representing x, y, h can be written down explicitly. Also for $i = m+1$, $xv_i = 0$ in the LHS of (iii), whereas the RHS is $(\lambda - m)v_m$. Since $v_m \neq 0$, $\lambda = m$, i.e. the weight of a maximal vector is a nonnegative integer. We call it the highest weight of V . Moreover each weight μ occurs with multiplicity one by (i). In particular, since V determines λ uniquely, the maximal vector v_0 is the only possible one in V .

Theorem 2.6. *Let V be an irreducible module for $L = \mathfrak{sl}(2, F)$.*

1. *Relative to h , V is the direct sum of weight spaces V_μ , $\mu = m, m-2, \dots, -(m-2), -m$ where $m+1 = \dim V$ and $\dim V_\mu = 1$ for each such μ .*
2. *V has up to nonzero scalar multiples a unique maximal vector, whose weight is m .*
3. *The action of L on V is given explicitly by the above formulas, if the basis is chosen in the prescribed fashion. In particular, there exists at most one irreducible L -module up to isomorphism of each possible dimension $m+1$, $m \geq 0$.*

pf. By the preceding paragraph.

Corollary 2.3. *Let V be any finite dimensional L -module, $L = \mathfrak{sl}(2, F)$. Then the eigenvalues of h on V are all integers and each occurs along with its negative. Moreover, in any decomposition of V into direct sum of irreducible submodules, the number of summands is precisely $\dim V_0 + \dim V_1$.*

pf. If $V = 0$ then nothing to prove. If $V \neq 0$, by the Weyl's theorem, we can write V as direct sum of irreducible submodules. Then by the previous theorem, the first part of the corollary follows. Each irreducible L -module has a unique occurrence of either the weight 0 or else the weight 1, but not both so the last part of the corollary follows.

Existence of the maximal vector

pf. Look at the subalgebra B spanned by h, x . It is a solvable subalgebra of $\mathfrak{gl}(V)$ with $\dim V < \infty$. Since $V \neq 0$, V contains a common eigenvector v for all the endomorphisms in B by the Lie's theorem ($[BB] = Fx$, $[FxFx] = 0$).

• **CLAIM : v is a maximal vector.**

pf. Since v is a common eigenvector for B , $\exists \lambda, \mu \in F$ s.t. $\begin{cases} hv = \lambda v \\ xv = \mu v \end{cases}$. Then $2xv = [hx]v = hxv - xhv = h(\mu v) - x(\lambda v) = \mu\lambda v - \lambda\mu v = 0$. Since $\text{char} F = 0$, $xv = 0$ so v is a maximal vector.

References

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